Ankush Sonawane

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EXPERIENCE

PROJECT ENGINEER , CDAC — APRIL 2024 – PRESENT

- Contributing to AI/ML model optimization, focusing on efficiency, scalability, and integration within real-world systems.
- Exploring large language models (LLMs) and AI frameworks for emerging applications.
- Supporting HPC performance analysis, including benchmarking of GPUs, CPUs, and scientific workloads.
- Developing internal tools and prototypes by combining AI/ML techniques with HPC infrastructure.
- Conducting technical sessions and creating awareness resources on AI, ML, and HPC for internal knowledge building.

PROJECTS

- **AI ATS System :** Developed an AI-powered Applicant Tracking System using vLLM and LangChain for resume parsing, candidate scoring, and recruitment insights with FastAPI backend and Streamlit interface.
- **RetrieverAI :** Built a RAG-based conversational AI system enabling users to query documents through a chat interface using LangChain, vLLM, and ChromaDB with a Streamlit frontend.
- **Retina-AV-Segmentation :** Implemented a deep learning pipeline for retinal artery-vein segmentation from fundus images using Multi-Class Attention U-Net with PyTorch and HPC-based distributed training.
- **Ancient Text Recognition (Bramhanetra) :** Recognized ancient Bramhi script using InceptionResNetV2, aiding in digitizing and preserving historical Indian inscriptions.
- **Agricultural Consulting System :** Web-based advisory system integrating ML models for crop & fertilizer recommendation and plant disease prediction using YOLO.

TECHNICAL CONTRIBUTION

- **DataScience :** Curated notebooks and implementations covering key data science concepts, algorithms, and workflows.
- **From Words to Understanding: How the Attention Mechanism Power Modern AI :** A comprehensive blog on the architecture of Transformers and its working.

EDUCATION

- **PG Diploma in Artificial Intelligence**, CDAC Innovation Park, Pune
- **B.E. in Computer Engineering**, Savitribai Phule Pune University, Pune

TECHNICAL SKILLSET

- **Machine Learning:** Model Selection, Training, Hyperparameter Tuning, Feature Engineering, Time Series Forecasting, Model Evaluation
- **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Model Development, Optimization, GPU Offloading, Multi-node/Multi-GPU Training
- **Natural Language Processing (NLP):** Text Preprocessing, Embeddings, Transformer-based Models, LLM Pipelines, Prompt Engineering, Retrieval-Augmented Generation (RAG)
- **Data Science:** Data Cleaning, Data Visualization, Exploratory Data Analysis (EDA), Hypothesis Testing, Statistical Inference
- **Programming & Scripting:** Python, C++
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Gradio, LangChain
- **Distributed Computing & HPC:** SLURM, Parallel Job Scheduling, Performance Benchmarking
- **Database Systems:** MySQL, Relational Database Design, SQL Querying, Data Integration, ChromaDB
- **Operating Systems:** Linux, Windows

SOFT SKILLS

Communication , Leadership , Teamwork and Collaboration , Learning Agility, Problem Solving